6.7 DEPLETION-TYPE MOSFET

As noted in the introduction, there are three types of FETs: JFETs, MOSFETs, and MESFETs. MOSFETs are further broken down into *depletion type* and *enhancement type*. The terms *depletion* and *enhancement* define their basic mode of operation; the name MOSFET stands for *m*etal–*o*xide–*s*emiconductor *f*ield-*e*ffect *t*ransistor. Since there are differences in the characteristics and operation of different types of MOSFET, they are covered in separate sections. In this section we examine the depletion-type MOSFET, which has characteristics similar to those of a JFET between cutoff and saturation at I_{DSS}, and also has the added feature of characteristics that extend into the region of opposite polarity for V_{GS}.

Basic Construction

The basic construction of the *n*-channel depletion-type MOSFET is provided in Fig. 6.24. A slab of *p*-type material is formed from a silicon base and is referred to as the *substrate*. It is the foundation on which the device is constructed. In some cases the substrate is internally connected to the source terminal. However, many discrete devices provide an additional terminal labeled *SS*, resulting in a four-terminal device, such as that in Fig. 6.24. The source and drain terminals are connected through metallic contacts to *n*-doped regions linked by an *n*-channel as shown in the figure. The gate is also connected to a metal contact surface but remains insulated from the *n*-channel by a very thin silicon dioxide (SiO_2) layer. SiO_2 is a type of insulator referred to as a *dielectric*, which sets up opposing (as indicated by the prefix *di-*) electric fields within the dielectric when exposed to an externally applied field. The fact that the SiO_2 layer is an insulating layer means that:

There is no direct electrical connection between the gate terminal and the channel of a MOSFET.

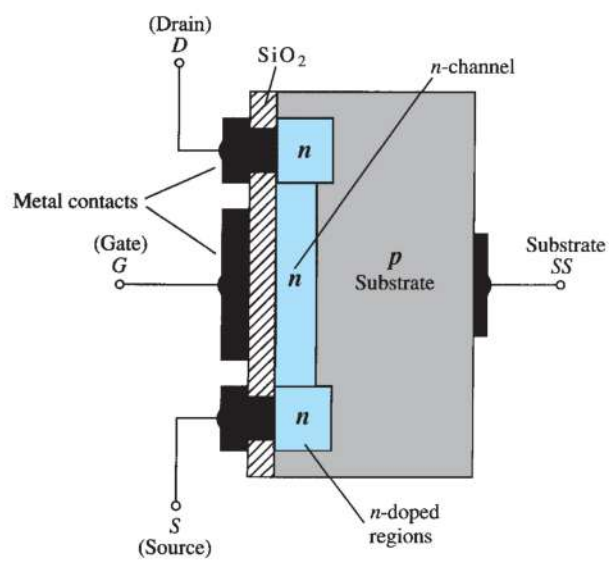

FIG. 6.24
n-Channel depletion-type MOSFET.

In addition:

It is the insulating layer of SiO_2 in the MOSFET construction that accounts for the very desirable high input impedance of the device.

In fact, the input resistance of a MOSFET is usually more than that of a typical JFET, even though the input impedance of most JFETs is sufficiently high for most applications. Because of the very high input impedance, the gate current I_G is essentially 0 A for dc-biased configurations.

The reason for the label metal–oxide–semiconductor FET is now fairly obvious: *metal* for the drain, source, and gate connections; *oxide* for the silicon dioxide insulating layer; and

semiconductor for the basic structure on which the *n*- and *p*-type regions are diffused. The insulating layer between the gate and the channel has resulted in another name for the device: *insulated-gate FET*, or *IGFET*, although this label is used less and less in the literature.

Basic Operation and Characteristics

In Fig. 6.25 the gate-to-source voltage is set to 0 V by the direct connection from one terminal to the other, and a voltage V_{DD} is applied across the drain-to-source terminals. The result is an attraction of the free electrons of the *n*-channel for the positive voltage at the drain. The result is a current similar to that flowing in the channel of the JFET. In fact, the resulting current with $V_{GS} = 0$ V continues to be labeled I_{DSS}, as shown in Fig. 6.26.

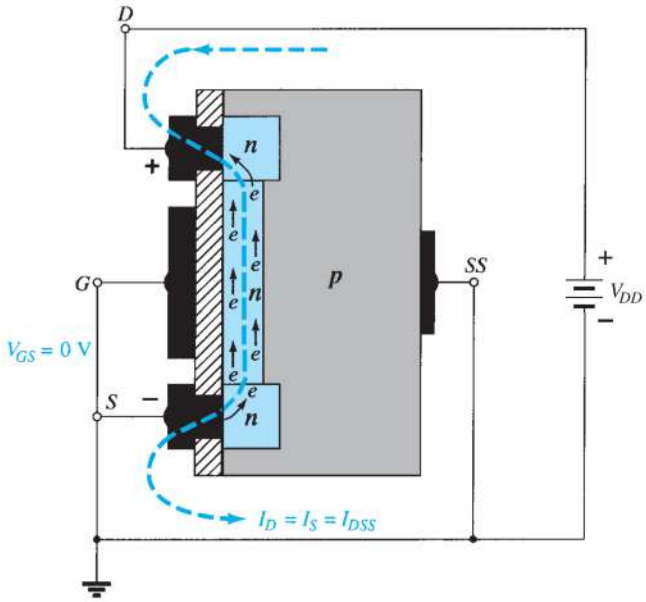

FIG. 6.25
n-Channel depletion-type MOSFET with $V_{GS} = 0$ V and applied voltage V_{DD}.

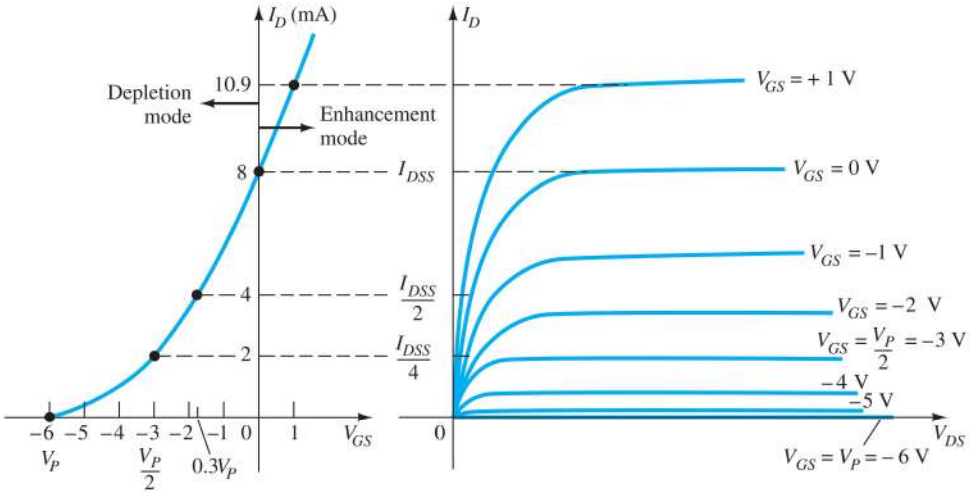

FIG. 6.26
Drain and transfer characteristics for an n-channel depletion-type MOSFET.

In Fig. 6.27, V_{GS} is set at a negative voltage such as −1 V. The negative potential at the gate will tend to pressure electrons toward the *p*-type substrate (like charges repel) and attract holes from the *p*-type substrate (opposite charges attract) as shown in Fig. 6.27. Depending on the

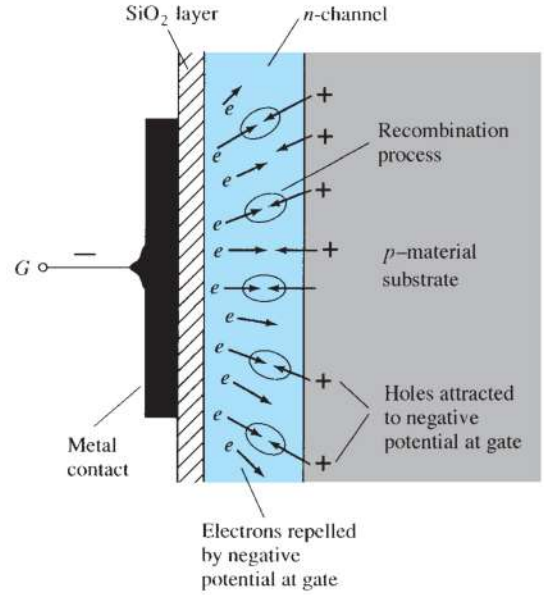

FIG. 6.27
Reduction in free carriers in a channel due to a negative potential at the gate terminal.

magnitude of the negative bias established by V_{GS}, a level of recombination between electrons and holes will occur that will reduce the number of free electrons in the *n*-channel available for conduction. The more negative the bias, the higher is the rate of recombination. The resulting level of drain current is therefore reduced with increasing negative bias for V_{GS}, as shown in Fig. 6.26 for $V_{GS} = -1$ V, -2 V, and so on, to the pinch-off level of -6 V. The resulting levels of drain current and the plotting of the transfer curve proceed exactly as described for the JFET.

For positive values of V_{GS}, the positive gate will draw additional electrons (free carriers) from the *p*-type substrate due to the reverse leakage current and establish new carriers through the collisions resulting between accelerating particles. As the gate-to-source voltage continues to increase in the positive direction, Fig. 6.26 reveals that the drain current will increase at a rapid rate for the reasons listed above. The vertical spacing between the $V_{GS} = 0$ V and $V_{GS} = +1$ V curves of Fig. 6.26 is a clear indication of how much the current has increased for the 1-V change in V_{GS}. Due to the rapid rise, the user must be aware of the maximum drain current rating since it could be exceeded with a positive gate voltage. That is, for the device of Fig. 6.26, the application of a voltage $V_{GS} = +4$ V would result in a drain current of 22.2 mA, which could possibly exceed the maximum rating (current or power) for the device. As revealed above, the application of a positive gate-to-source voltage has "enhanced" the level of free carriers in the channel compared to that encountered with $V_{GS} = 0$ V. For this reason the region of positive gate voltages on the drain or transfer characteristics is often referred to as the *enhancement region*, with the region between cutoff and the saturation level of I_{DSS} referred to as the *depletion region.*

It is particularly interesting and helpful that Shockley's equation will continue to be applicable for the depletion-type MOSFET characteristics in both the depletion and enhancement regions. For both regions, it is simply necessary that the proper sign be included with V_{GS} in the equation and the sign be carefully monitored in the mathematical operations.

EXAMPLE 6.3 Sketch the transfer characteristics for an *n*-channel depletion-type MOSFET with $I_{DSS} = 10$ mA and $V_P = -4$ V.

Solution:

$$\text{At } V_{GS} = 0\text{ V}, \quad I_D = I_{DSS} = 10\text{ mA}$$

$$V_{GS} = V_P = -4\text{ V}, \quad I_D = 0\text{ mA}$$

$$V_{GS} = \frac{V_P}{2} = \frac{-4\text{ V}}{2} = -2\text{ V}, \quad I_D = \frac{I_{DSS}}{4} = \frac{10\text{ mA}}{4} = 2.5\text{ mA}$$

and at $I_D = \dfrac{I_{DSS}}{2}$,

$$V_{GS} = 0.3V_P = 0.3(-4\text{ V}) = -1.2\text{ V}$$

all of which appear in Fig. 6.28.

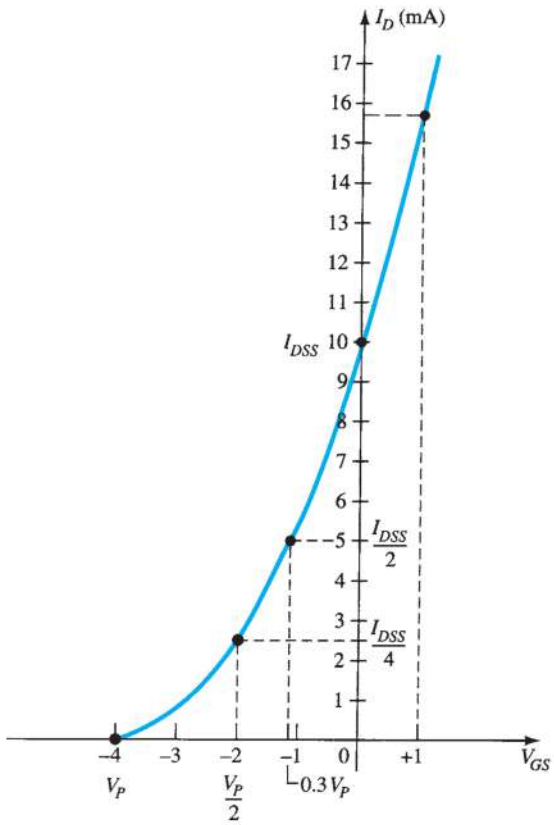

FIG. 6.28
Transfer characteristics for an n-channel depletion-type MOSFET with I_{DSS} = 10 mA and V_P = −4 V.

Before plotting the positive region of V_{GS}, keep in mind that I_D increases very rapidly with increasing positive values of V_{GS}. In other words, be conservative with the choice of values to be substituted into Shockley's equation. In this case, we try +1 V as follows:

$$I_D = I_{DSS}\left(1 - \frac{V_{GS}}{V_P}\right)^2$$

$$= (10\text{ mA})\left(1 - \frac{+1\text{ V}}{-4\text{ V}}\right)^2 = (10\text{ mA})(1 + 0.25)^2 = (10\text{ mA})(1.5625)$$

$$\cong 15.63\text{ mA}$$

which is sufficiently high to finish the plot.

p-Channel Depletion-Type MOSFET

The construction of a *p*-channel depletion-type MOSFET is exactly the reverse of that appearing in Fig. 6.24. That is, there is now an *n*-type substrate and a *p*-type channel, as shown in Fig. 6.29a. The terminals remain as identified, but all the voltage polarities and the current directions are reversed, as shown in the same figure. The drain characteristics would appear exactly as in Fig. 6.26, but with V_{DS} having negative values, I_D having positive values as indicated (since the defined direction is now reversed), and V_{GS} having the opposite polarities as shown in Fig. 6.29c. The reversal in V_{GS} will result in a mirror image (about the I_D axis) for the transfer characteristics as shown in Fig. 6.29b. In other words,

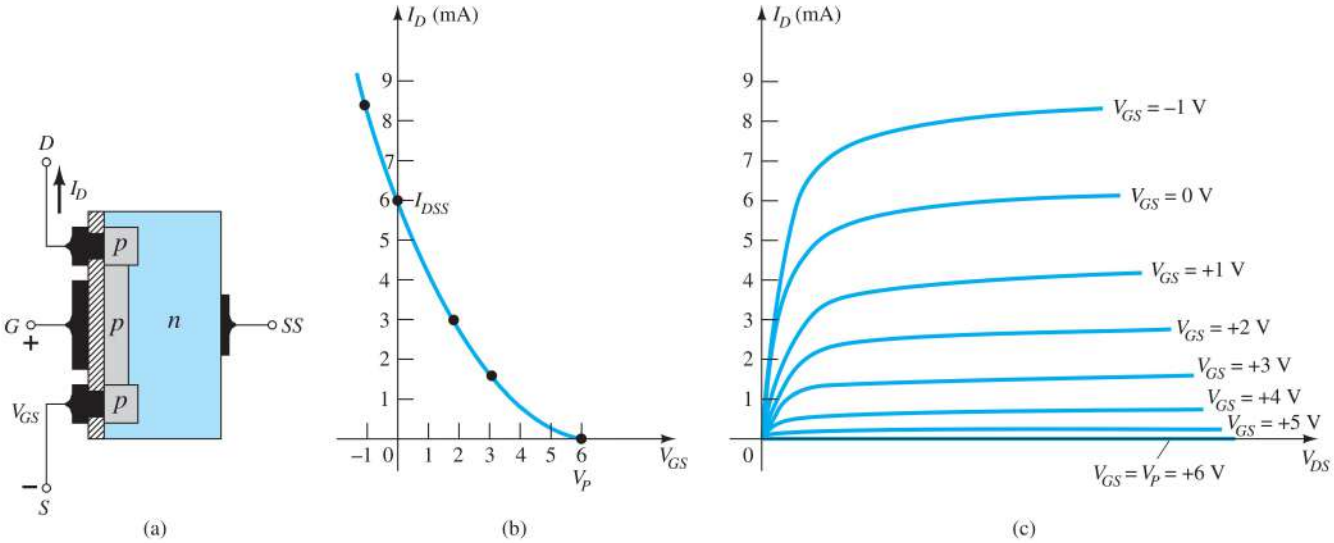

FIG. 6.29
p-Channel depletion-type MOSFET with $I_{DSS} = 6$ mA and $V_P = +6$ V.

the drain current will increase from cutoff at $V_{GS} = V_P$ in the positive V_{GS} region to I_{DSS} and then continue to increase for increasingly negative values of V_{GS}. Shockley's equation is still applicable and requires simply placing the correct sign for both V_{GS} and V_P in the equation.

Symbols, Specification Sheets, and Case Construction

The graphic symbols for an *n*- and *p*-channel depletion-type MOSFET are provided in Fig. 6.30. Note how the symbols chosen try to reflect the actual construction of the device. The lack of a direct connection (due to the gate insulation) between the gate and the channel is represented by a space between the gate and the other terminals of the symbol. The vertical line representing the channel is connected between the drain and the source and is "supported" by the substrate. Two symbols are provided for each type of channel to reflect the fact that in some cases the substrate is externally available, whereas in others it is not. For most of the analysis to follow in Chapter 7, the substrate and the source will be connected and the lower symbols will be employed.

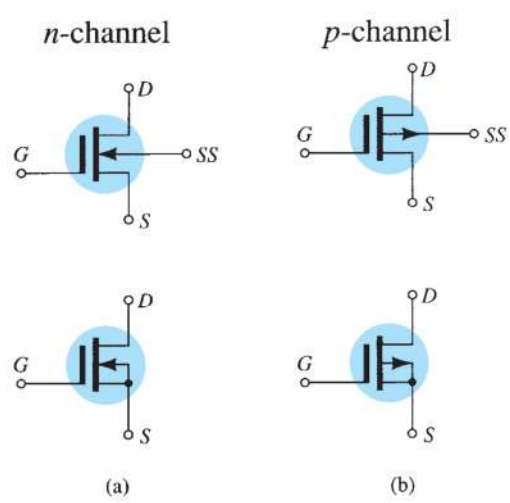

FIG. 6.30
Graphic symbols for: (a) n-channel depletion-type MOSFETs and (b) p-channel depletion-type MOSFETs.

6.8 ENHANCEMENT-TYPE MOSFET

Although there are some similarities in construction and mode of operation between depletion-type and enhancement-type MOSFETs, the characteristics of the enhancement-type MOSFET are quite different from anything obtained thus far. The transfer curve is not defined by Shockley's equation, and the drain current is now cut off until the gate-to-source voltage reaches a specific magnitude. In particular, current control in an *n*-channel device is now effected by a positive gate-to-source voltage rather than the range of negative voltages encountered for *n*-channel JFETs and *n*-channel depletion-type MOSFETs.

Basic Construction

The basic construction of the *n*-channel enhancement-type MOSFET is provided in Fig. 6.32. A slab of *p*-type material is formed from a silicon base and is again referred to as the substrate. As with the depletion-type MOSFET, the substrate is sometimes internally connected to the source terminal, whereas in other cases a fourth lead (labeled SS) is made available for external control of its potential level. The source and drain terminals are again connected through metallic contacts to *n*-doped regions, but note in Fig. 6.32 the absence of a channel between the two *n*-doped regions. This is the primary difference between the construction of depletion-type and enhancement-type MOSFETs—the absence of a channel as a constructed component of the device. The SiO_2 layer is still present to isolate the gate metallic platform from the region between the drain and source, but now it is simply separated from a section of the *p*-type material. In summary, therefore, the construction of an enhancement-type MOSFET is quite similar to that of the depletion-type MOSFET, except for the absence of a channel between the drain and source terminals.

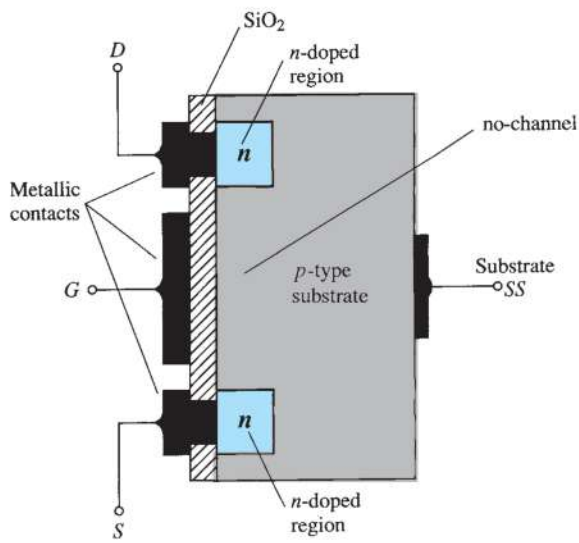

FIG. 6.32
n-Channel enhancement-type MOSFET.

Basic Operation and Characteristics

If V_{GS} is set at 0 V and a voltage applied between the drain and the source of the device of Fig. 6.32, the absence of an *n*-channel (with its generous number of free carriers) will result in a current of effectively 0 A—quite different from the depletion-type MOSFET and JFET, where $I_D = I_{DSS}$. It is not sufficient to have a large accumulation of carriers (electrons) at the drain and the source (due to the *n*-doped regions) if a path fails to exist between the two. With V_{DS} some positive voltage, V_{GS} at 0 V, and terminal *SS* directly connected to the source, there are in fact two reverse-biased *p–n* junctions between the *n*-doped regions and the *p*-substrate to oppose any significant flow between drain and source.

In Fig. 6.33, both V_{DS} and V_{GS} have been set at some positive voltage greater than 0 V, establishing the drain and the gate at a positive potential with respect to the source.

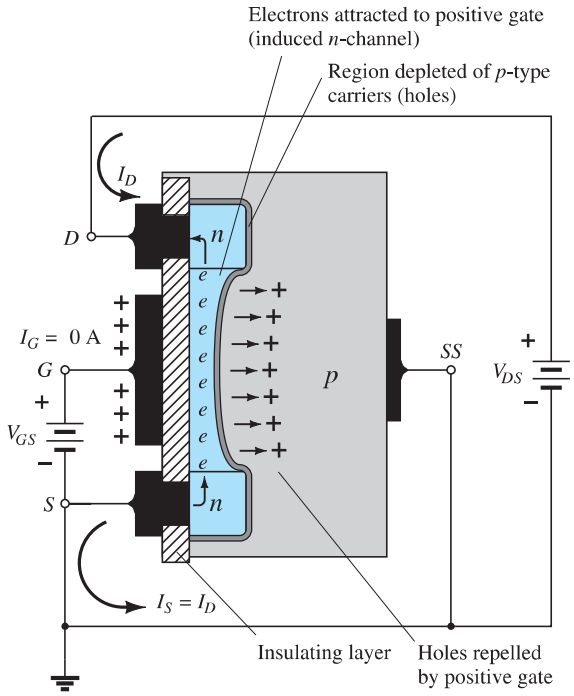

FIG. 6.33
Channel formation in the n-channel enhancement-type MOSFET.

The positive potential at the gate will pressure the holes (since like charges repel) in the *p*-substrate along the edge of the SiO_2 layer to leave the area and enter deeper regions of the *p*-substrate, as shown in the figure. The result is a depletion region near the SiO_2 insulating layer void of holes. However, the electrons in the *p*-substrate (the minority carriers of the material) will be attracted to the positive gate and accumulate in the region near the surface of the SiO_2 layer. The SiO_2 layer and its insulating qualities will prevent the negative carriers from being absorbed at the gate terminal. As V_{GS} increases in magnitude, the concentration of electrons near the SiO_2 surface increases until eventually the induced *n*-type region can support a measurable flow between drain and source. The level of V_{GS} that results in the significant increase in drain current is called the *threshold voltage* and is given the symbol V_T. On specification sheets it is referred to as $V_{GS(\text{Th})}$, although V_T is less unwieldy and will be used in the analysis to follow. Since the channel is nonexistent with $V_{GS} = 0$ V and "enhanced" by the application of a positive gate-to-source voltage, this type of MOSFET is called an *enhancement-type MOSFET*. Both depletion- and enhancement-type MOSFETs have enhancement-type regions, but the label was applied to the latter since it is its only mode of operation.

As V_{GS} is increased beyond the threshold level, the density of free carriers in the induced channel will increase, resulting in an increased level of drain current. However, if we hold V_{GS} constant and increase the level of V_{DS}, the drain current will eventually reach a saturation level as occurred for the JFET and depletion-type MOSFET. The leveling off of I_D is due to a pinching-off process depicted by the narrower channel at the drain end of the induced channel as shown in Fig. 6.34. Applying Kirchhoff's voltage law to the terminal voltages of the MOSFET of Fig. 6.34, we find that

$$\boxed{V_{DG} = V_{DS} - V_{GS}} \tag{6.13}$$

If V_{GS} is held fixed at some value such as 8 V and V_{DS} is increased from 2 V to 5 V, the voltage V_{DG} [by Eq. (6.13)] will increase from −6 V to −3 V and the gate will become less and less positive with respect to the drain. This reduction in gate-to-drain voltage will in turn reduce the attractive forces for free carriers (electrons) in this region of the induced channel, causing a reduction in the effective channel width. Eventually, the channel will be reduced to the point of pinch-off and a saturation condition will be established as described

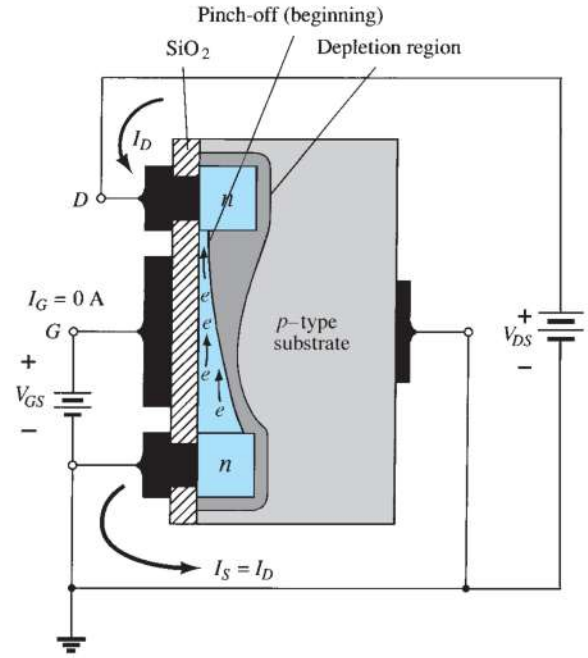

FIG. 6.34
Change in channel and depletion region with increasing level of V_{DS} for a fixed value of V_{GS}.

earlier for the JFET and depletion-type MOSFET. In other words, any further increase in V_{DS} at the fixed value of V_{GS} will not affect the saturation level of I_D until breakdown conditions are encountered.

The drain characteristics of Fig. 6.35 reveal that for the device of Fig. 6.34 with $V_{GS} = 8$ V, saturation occurs at a level of $V_{DS} = 6$ V. In fact, the saturation level for V_{DS} is related to the level of applied V_{GS} by

$$V_{DS_{sat}} = V_{GS} - V_T \tag{6.14}$$

Obviously, therefore, for a fixed value of V_T, the higher the level of V_{GS}, the greater is the saturation level for V_{DS}, as shown in Fig. 6.34 by the locus of saturation levels.

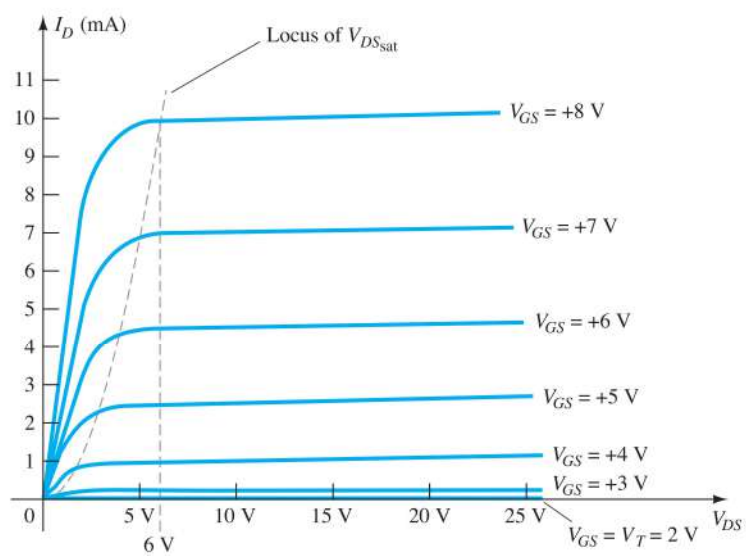

FIG. 6.35
Drain characteristics of an n-channel enhancement-type MOSFET with $V_T = 2$ V and $k = 0.278 \times 10^{-3}$ A/V^2.

For the characteristics of Fig. 6.34, the level of V_T is 2 V, as revealed by the fact that the drain current has dropped to 0 mA. In general, therefore:

For values of V_{GS} less than the threshold level, the drain current of an enhancement-type MOSFET is 0 mA.

Figure 6.35 clearly reveals that as the level of V_{GS} increases from V_T to 8 V, the resulting saturation level for I_D also increases from a level of 0 mA to 10 mA. In addition, it is quite noticeable that the spacing between the levels of V_{GS} increases as the magnitude of V_{GS} increases, resulting in ever-increasing increments in drain current.

For levels of $V_{GS} > V_T$, the drain current is related to the applied gate-to-source voltage by the following nonlinear relationship:

$$\boxed{I_D = k(V_{GS} - V_T)^2} \tag{6.15}$$

Again, it is the squared term that results in the nonlinear (curved) relationship between I_D and V_{GS}. The k term is a constant that is a function of the construction of the device. The value of k can be determined from the following equation [derived from Eq. (6.15)], where $I_{D(\text{on})}$ and $V_{GS(\text{on})}$ are the values for each at a particular point on the characteristics of the device.

$$\boxed{k = \frac{I_{D(\text{on})}}{(V_{GS(\text{on})} - V_T)^2}} \tag{6.16}$$

Substituting $I_{D(\text{on})} = 10$ mA when $V_{GS(\text{on})} = 8$ V from the characteristics of Fig. 6.35 yields

$$k = \frac{10\text{ mA}}{(8\text{ V} - 2\text{ V})^2} = \frac{10\text{ mA}}{(6\text{ V})^2} = \frac{10\text{ mA}}{36\text{ V}^2}$$
$$= \mathbf{0.278 \times 10^{-3}\ A/V^2}$$

and a general equation for I_D for the characteristics of Fig. 6.35 results in

$$I_D = 0.278 \times 10^{-3}(V_{GS} - 2\text{ V})^2$$

Substituting $V_{GS} = 4$ V, we find that

$$I_D = 0.278 \times 10^{-3}(4\text{ V} - 2\text{ V})^2 = 0.278 \times 10^{-3}(2)^2$$
$$= 0.278 \times 10^{-3}(4) = \mathbf{1.11\ mA}$$

as verified by Fig. 6.35. At $V_{GS} = V_T$, the squared term is 0, and $I_D = 0$ mA.

For the dc analysis of enhancement-type MOSFETs to appear in Chapter 7, the transfer characteristics will again be the characteristics to be employed in the graphical solution. In Fig. 6.36, the drain and transfer characteristics have been set side by side to describe the

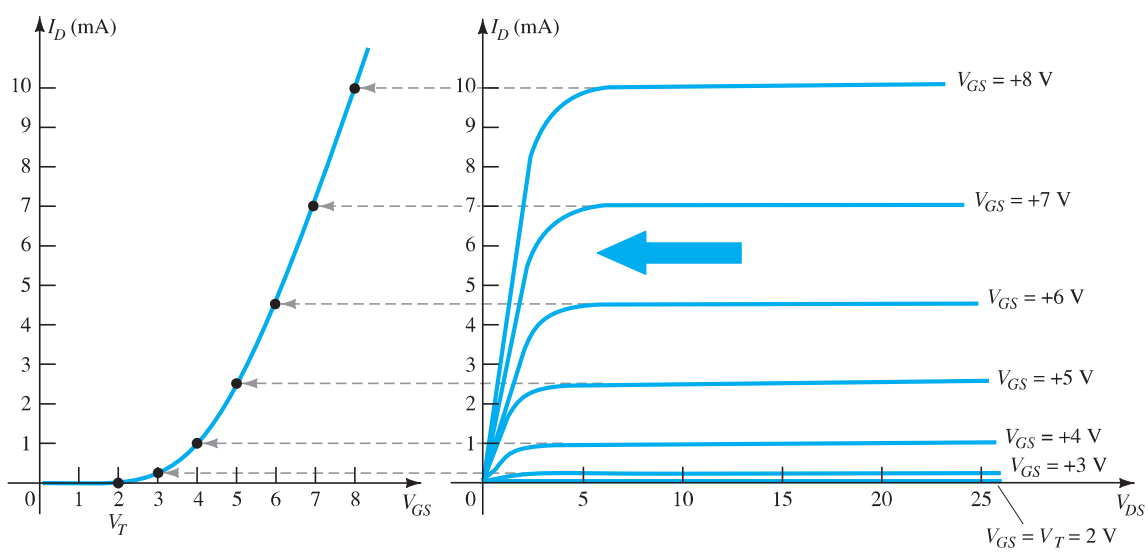

FIG. 6.36

Sketching the transfer characteristics for an n-channel enhancement-type MOSFET from the drain characteristics.

transfer process from one to the other. Essentially, it proceeds as introduced earlier for the JFET and depletion-type MOSFETs. In this case, however, it must be remembered that the drain current is 0 mA for $V_{GS} \leq V_T$. As V_{GS} is increased beyond V_T, the drain current I_D will begin to flow at an increasing rate in accordance with Eq. (6.15). Note that in defining the points on the transfer characteristics from the drain characteristics, only the saturation levels are employed, thereby limiting the region of operation to levels of V_{DS} greater than the saturation levels as defined by Eq. (6.14).

The transfer curve of Fig. 6.36 is certainly quite different from those obtained earlier. For an *n*-channel (induced) device, it is now totally in the positive V_{GS} region and does not rise until $V_{GS} = V_T$. The question now surfaces as to how to plot the transfer characteristics given the levels of k and V_T as included below for a particular MOSFET:

$$I_D = 0.5 \times 10^{-3}(V_{GS} - 4\text{ V})^2$$

First, a horizontal line is drawn at $I_D = 0$ mA from $V_{GS} = 0$ V to $V_{GS} = 4$ V as shown in Fig. 6.37a. Next, a level of V_{GS} greater than V_T such as 5 V is chosen and substituted into Eq. (6.15) to determine the resulting level of I_D as follows:

$$\begin{aligned} I_D &= 0.5 \times 10^{-3}(V_{GS} - 4\text{ V})^2 \\ &= 0.5 \times 10^{-3}(5\text{ V} - 4\text{ V})^2 = 0.5 \times 10^{-3}(1)^2 \\ &= \mathbf{0.5\ mA} \end{aligned}$$

and a point on the plot is obtained as shown in Fig. 6.37b. Finally, additional levels of V_{GS} are chosen and the resulting levels of I_D obtained. In particular, at $V_{GS} = 6$, 7, and 8 V, the level of I_D is 2, 4.5, and 8 mA, respectively, as shown on the resulting plot of Fig. 6.37c.

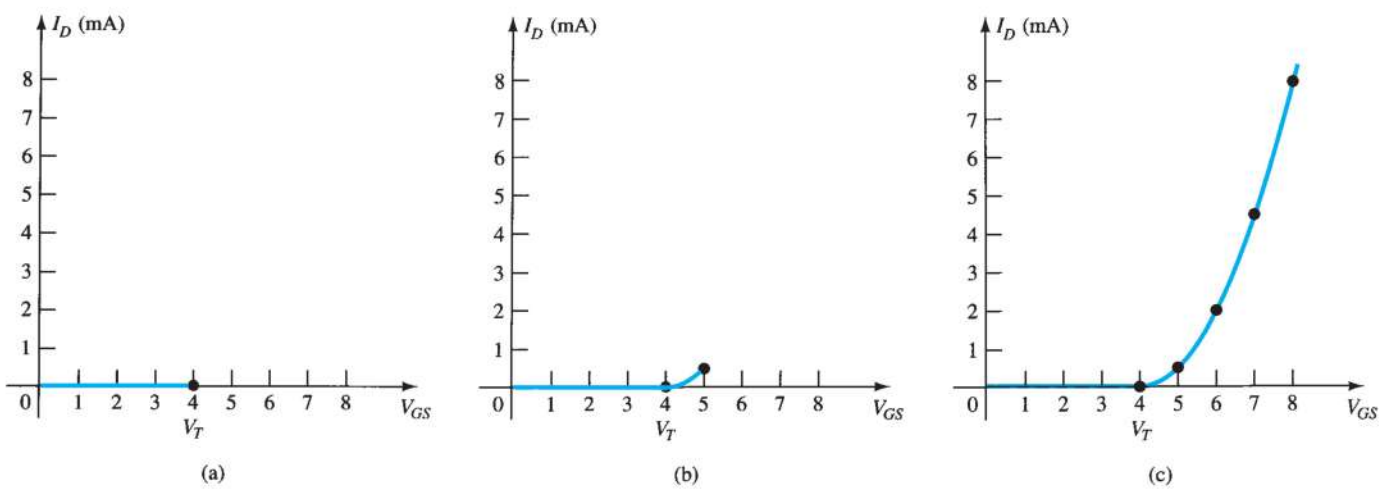

FIG. 6.37
Plotting the transfer characteristics of an n-channel enhancement-type MOSFET with $k = 0.5 \times 10^{-3}$ A/V^2 and $V_T = 4$ V.

p-Channel Enhancement-Type MOSFETs

The construction of a *p*-channel enhancement-type MOSFET is exactly the reverse of that appearing in Fig. 6.32, as shown in Fig. 6.38a. That is, there is now an *n*-type substrate and *p*-doped regions under the drain and source connections. The terminals remain as identified, but all the voltage polarities and the current directions are reversed. The drain characteristics will appear as shown in Fig. 6.38c, with increasing levels of current resulting from increasingly negative values of V_{GS}. The transfer characteristics of Fig. 6.38b will be the mirror image (about the I_D axis) of the transfer curve of Fig. 6.36, with I_D increasing with increasingly negative values of V_{GS} beyond V_T, as shown in Fig. 6.38c. Equations (6.13) through (6.16) are equally applicable to *p*-channel devices.

Symbols, Specification Sheets, and Case Construction

The graphic symbols for the *n*- and *p*-channel enhancement-type MOSFETs are provided as Fig. 6.39. Again note how the symbols try to reflect the actual construction of

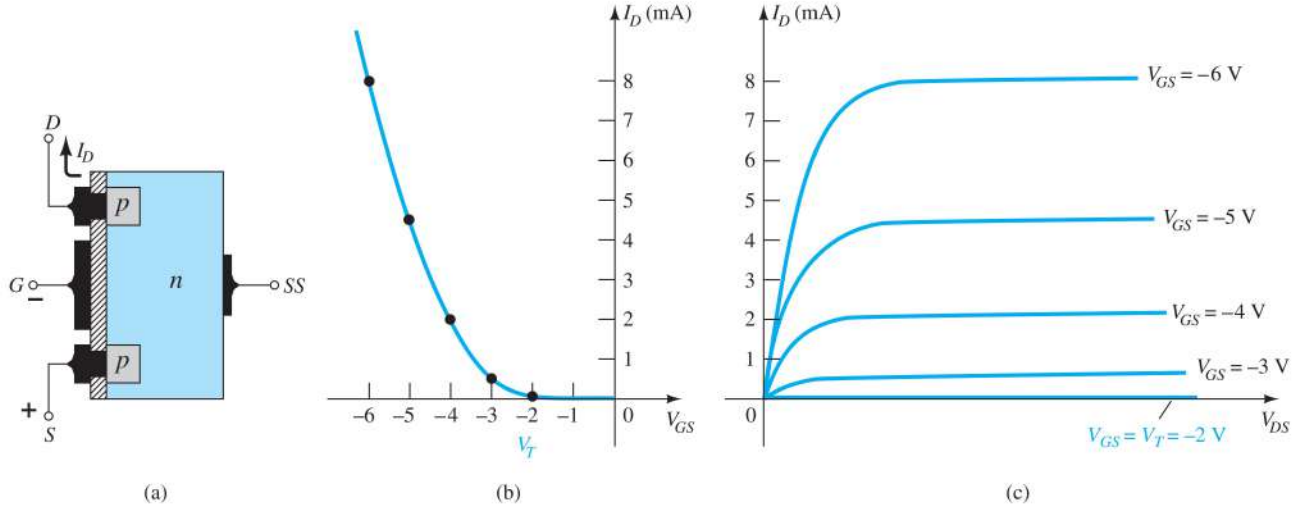

FIG. 6.38
p-Channel enhancement-type MOSFET with $V_T = 2$ V and $k = 0.5 \times 10^{-3}$ A/V^2.

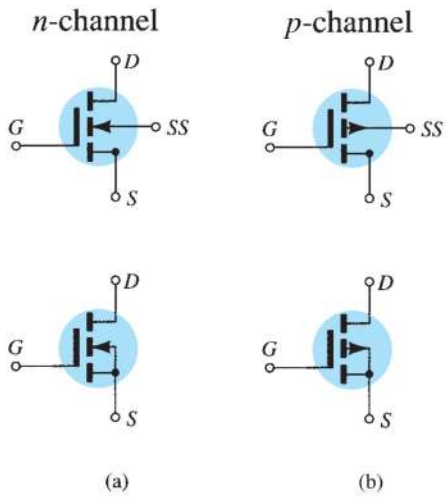

FIG. 6.39
Symbols for: (a) n-channel enhancement-type MOSFETs and (b) p-channel enhancement-type MOSFETs.

the device. The dashed line between drain and source is chosen to reflect the fact that a channel does not exist between the two under no-bias conditions. It is, in fact, the only difference between the symbols for the depletion-type and enhancement-type MOSFETs.

The specification sheet for a Motorola *n*-channel enhancement-type MOSFET is provided as Fig. 6.40. The case construction and the terminal identification are provided next to the maximum ratings, which now include a maximum drain current of 30 mA dc. The specification sheet provides the level of I_{DSS} under "off" conditions, which is now simply 10 nA dc (at $V_{DS} = 10$ V and $V_{GS} = 0$ V), compared to the milliampere range for the JFET and the depletion-type MOSFET. The threshold voltage is specified as $V_{GS(\text{Th})}$ and has a range of 1 to 5 V dc, depending on the device employed. Rather than provide a range of k in Eq. (6.15), a typical level of $I_{D(\text{on})}$ (3 mA in this case) is specified at a particular level of $V_{GS(\text{on})}$ (10 V for the specified I_D level). In other words, when $V_{GS} = 10$ V, $I_D = 3$ mA. The given levels of $V_{GS(\text{Th})}$, $I_{D(\text{on})}$, and $V_{GS(\text{on})}$ permit a determination of k from Eq. (6.16) and a writing of the general equation for the transfer characteristics. The handling requirements of MOSFETs are reviewed in Section 6.9.